

CARLOS FREITAS

Computer Science Engineer

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SUMMARY

Computer Science Engineer with a broad technical foundation across software development, systems, databases, networking, cloud technologies and data analytics, combined with over 11 years of professional experience in a highly technical and mission-critical environment.

I bring strong problem-solving and troubleshooting skills, a multidisciplinary mindset, and extensive experience working with complex systems and leading technical teams. My Computer Science background and hands-on projects have allowed me to work with different programming languages, technologies and infrastructure environments.

Motivated to build a career in the IT industry, contribute to challenging projects and continuously develop my skills as an engineer.

WORK EXPERIENCE

Senior Avionics Technician | Technical Inspector | Team Leader

Sep 2014 – Present

Portuguese Air Force – Beja, Portugal

- Leading a team of 8 avionics technicians, coordinating the maintenance, inspection and troubleshooting of avionics, electrical and electronic systems on Leonardo AW119 helicopters.
- Performing software updates, configuration and calibration of Garmin G1000H integrated avionics systems.
- Testing and troubleshooting complex communication, navigation, electrical and electronic systems, using technical documentation and structured fault-isolation procedures.
- Working in a mission-critical environment requiring high levels of technical accuracy, responsibility and teamwork.

IT Support Technician

May 2014 – Aug 2014

CGI – Lisbon, Portugal

- Delivered multilingual B2B technical support.
- Resolved software and application issues.
- Managed incidents through CRM and ticketing systems.

Mobile Network Operations Technician

May 2013 – Apr 2014

Nokia Solutions and Networks – Lisbon, Portugal

- Monitored mobile telecommunications network performance and availability.
- Performed remote troubleshooting and fault analysis across the Spanish mobile network.
- Opened, tracked and managed technical incidents through CRM and ticketing systems.
- Collaborated with technical teams to ensure network service continuity and rapid incident resolution.

EDUCATION

Bachelor's Degree in Computer Science Engineering

Sep 2023 – Jun 2026

- Polytechnic University of Beja (UPBeja)

Air Force Sergeant Course – Aircraft Electrical & Avionics

Dec 2017 – Dec 2018

- Portuguese Air Force Training and Technical Centre (CFMTFA)

Air Force Airman Course – Aircraft Electrical & Avionics

Nov 2014 – Sep 2015

- Portuguese Air Force Training and Technical Centre (CFMTFA)

EQF Level 4 Vocational Diploma in Industrial Maintenance/Electromechanics Nov 2007 – Sep 2010

- Professional School of Almada (EPA)

PROJECTS

HELINK — Helicopter Flight Data Analysis & Maintenance Support

Independent Project · 2026–Present

Python · PySide6 · SQLite · Pandas · Matplotlib · Git

- Desktop application for helicopter flight data analysis and preventive maintenance support, using flight data exported from Garmin G1000H avionics systems.
- Processes and synchronizes aircraft telemetry data, providing interactive visualization, alerts and exceedance analysis, aircraft and flight management, offline mapping and automated PDF maintenance reports.
- Built as an offline-first solution, combining data processing, local persistence and visualization to support technical analysis and maintenance decision-making.

StreamBolt TV — React Full-Stack Application

Academic Project

React · Node.js · Express · MongoDB · Vite · REST APIs

- Full-stack movie management platform featuring authentication, role-based authorization, CRUD operations, advanced search, moderated comments and external REST API integration.

Swim Training Manager — Android Application

Academic Project

Kotlin · Android · Jetpack Compose · Material Design 3

- Mobile application for managing swimming training sessions, athletes and clubs, with lap timing, workout history and automatic calculation of performance metrics.

SKILLS

Software Development

- Python · Java · JavaScript · Kotlin · SQL · REST APIs · React · Laravel · ASP.NET

Systems & Infrastructure

- Linux · Windows · Network Administration (Cisco & MikroTik) · Docker

Data & Analytics

- Power Query · Power BI · Data Analysis (Pandas) · Machine Learning (Scikit-learn)

Electronics & Electrical Engineering

- AC/DC Electrical Systems · Analog & Digital Electronics · Automotive Electrical & Electronic Systems · Instrumentation & Sensors · Fault Diagnosis

Avionics & Aerospace

- Garmin G1000H NXi · Aircraft Avionics · Aircraft Electrical Systems · Navigation Systems

Communications Systems

- Radio Communications · VHF/UHF Systems · Aircraft Communication Systems

Mechanical Systems

- Mechanical Systems · Electromechanical Systems · Automotive Mechanics

Leadership & Management

- Team Leadership · Personnel Management · Technical Decision Making · Technical Training · Planning & Coordination · Technical Documentation

LANGUAGES

Portuguese

- Native

Spanish

- Native

English

- B2